MECHANICAL ENGINEERING PORTFOLIO

ASTRA

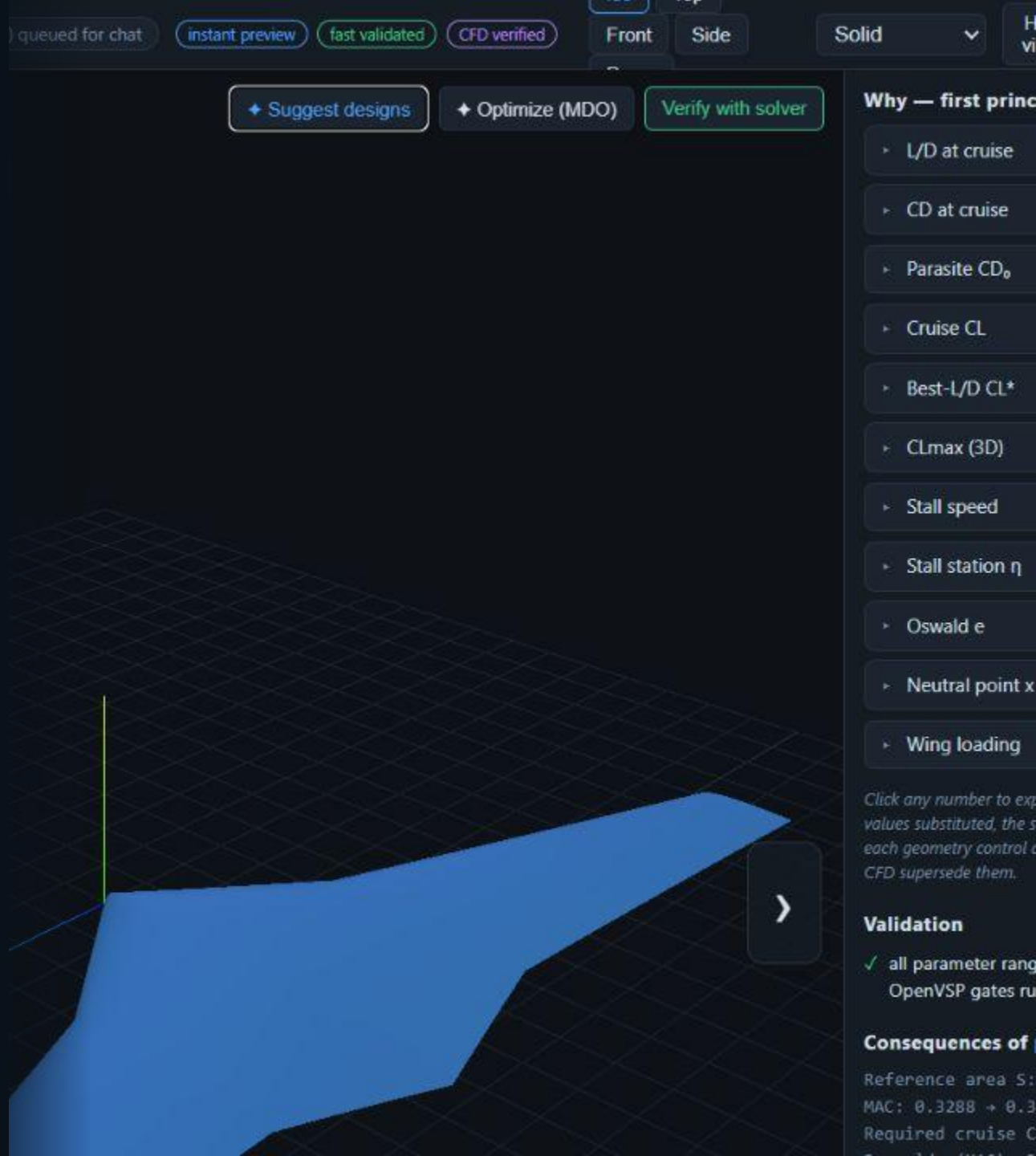
Mission intent -> parametric geometry -> mesh -> CFD evidence.

A reviewable blended-wing-body drone design workflow built around explicit assumptions, deterministic gates and honest result quality.

Lindon Vaz

Mechanical Design and CAE

github.com/L-Vaz/Astra



CHALLENGE AND OWNERSHIP

A mission brief becomes a reviewable engineering problem

Astra connects requirements, geometry and simulation without hiding the assumptions between them.

Concept design often fragments the requirement, CAD model and analysis case. Astra keeps those stages linked, versioned and inspectable.

INHERITED FOAM-AGENT FOUNDATION

- LangGraph CFD workflow
- Case planning, input writing and execution
- Review and correction loop
- RAG-backed OpenFOAM configuration
- Post-processing and visualization foundation

ASTRA EXTENSION

- BWB requirement schema and inverse sizing
- Parametric OpenVSP geometry
- Concept candidates and cockpit
- VSPAERO screening and deterministic Gmsh farfield
- Versioned sessions, gates and result-quality verdict

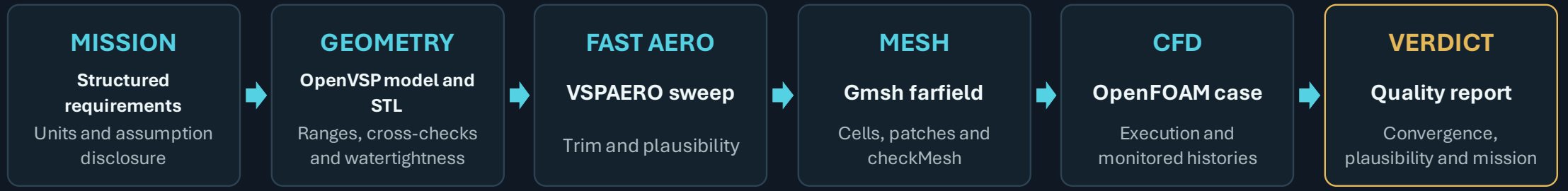
Aerodynamics-first conceptual design. Structures, propulsion, manufacturing and flight validation remain outside the present scope.

Source: README.md

DESIGN SYSTEM

One traceable loop links intent to evidence

Every stage creates an artifact, applies a gate and invalidates downstream evidence when the design changes.



REVISION BEHAVIOUR

Upstream revisions preserve history but invalidate downstream results until those stages are rerun.

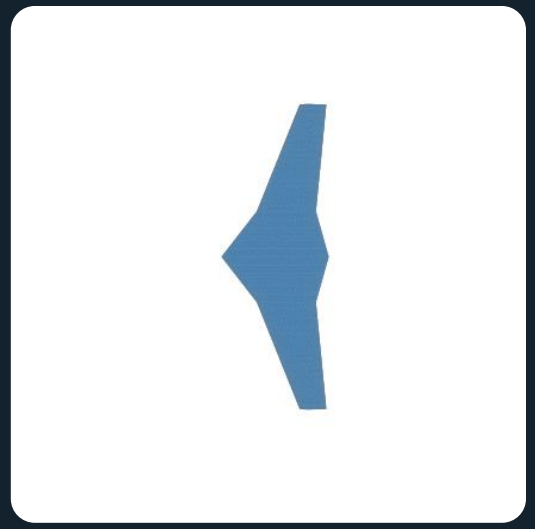
The engineering record stays reviewable: what changed, what became stale and what must be rerun remain explicit.

Source: Astra engineering guidance

REQUIREMENT-DRIVEN DESIGN

The concept is sized from forces, not styling

Steady level flight requires lift to balance weight; the requirement solver closes the planform around that physical constraint.



FIRST-PRINCIPLES SIZING

L = W; q = 0.5 rho V^2; S = W / (q CL)

Lift balances weight in steady level flight. Dynamic pressure converts the mission condition into the wing area required at the chosen lift coefficient.

RECORDED GOLDEN CASE [C]

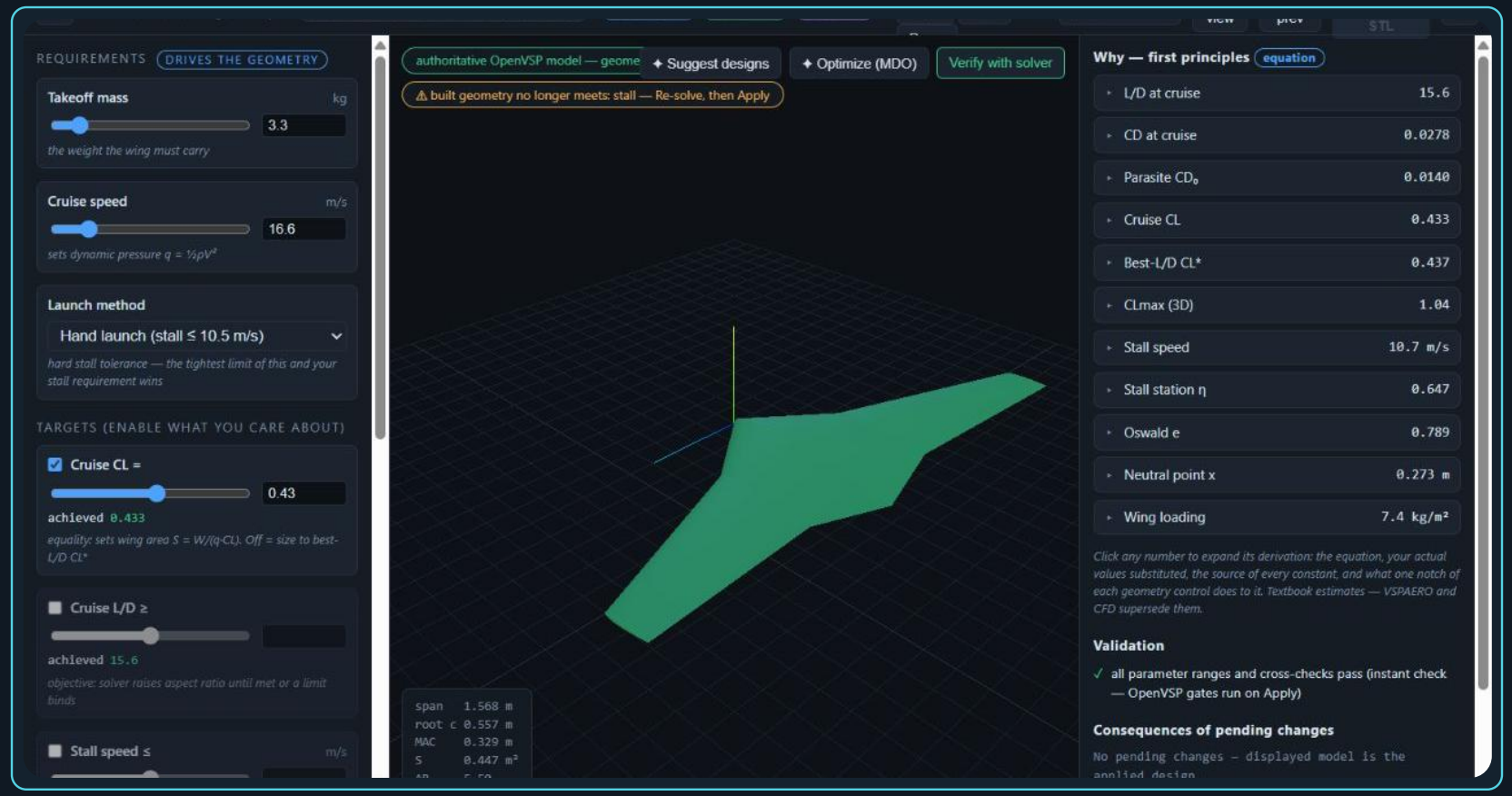
Mission payload	Cruise condition	Reference area	Span / aspect ratio
2 kg [C]	25 m/s at 100 m [C]	0.952 m2 [C]	2.29 m / 5.5 [C]

The interface distinguishes the value requested, the value in force and the bound that forced any change. Constraint closure remains visible instead of silently rewriting the brief.

Source: recorded verification report [C]

Design variables stay connected to their consequences

The cockpit exposes geometry, constraints and first-principles explanations while server-side validation remains authoritative.



WHAT THE INTERFACE EXPOSES

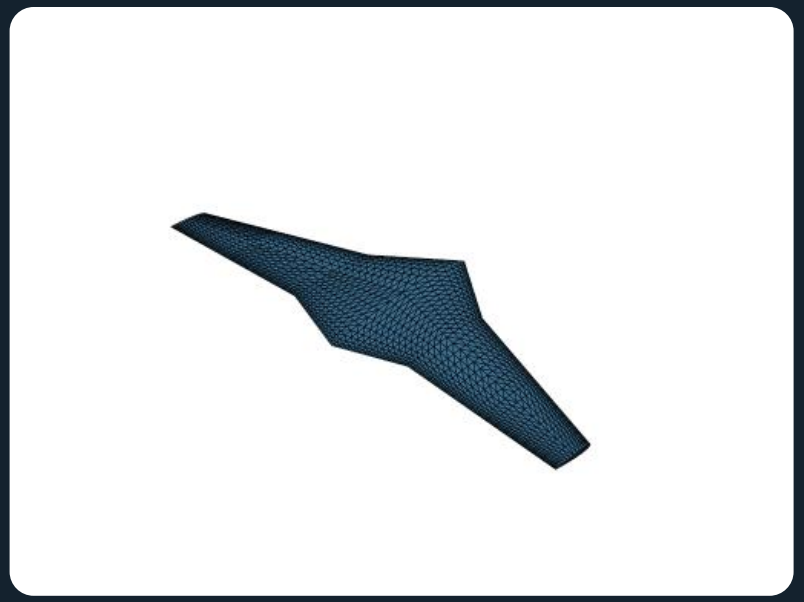
- Requirement controls show targets, achieved values and active bounds.
- The WebGL planform is an instant preview; Apply rebuilds the authoritative OpenVSP model.
- Existing candidates can be compared without overwriting the accepted geometry.
- Equation estimates remain labelled separately from VSPAERO and CFD evidence.

REAL EXISTING SESSION CAPTURE [C]

Source: Astra live code

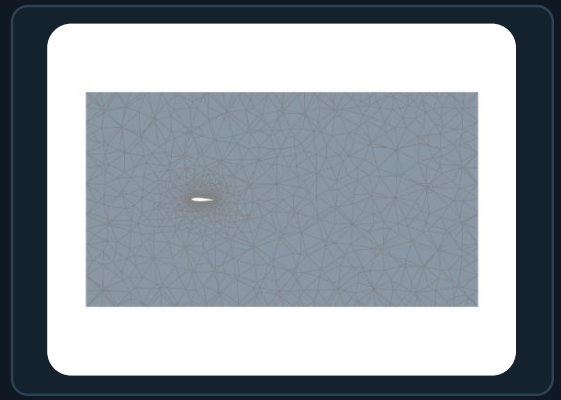
Deterministic mesh gates protect the solver handoff

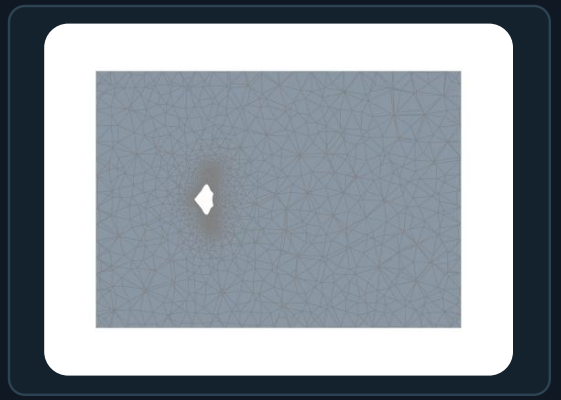
The validated OpenVSP surface is retained as the boundary while the farfield volume and patch types are generated and checked deterministically.



MEASURED MESH-QUALITY CHANGE [C]

Netgen optimization reduced average non-orthogonality from 22.6 to 17.9 and maximum skewness from 2.6 to 1.5 [C].





DETERMINISTIC HANDOFF

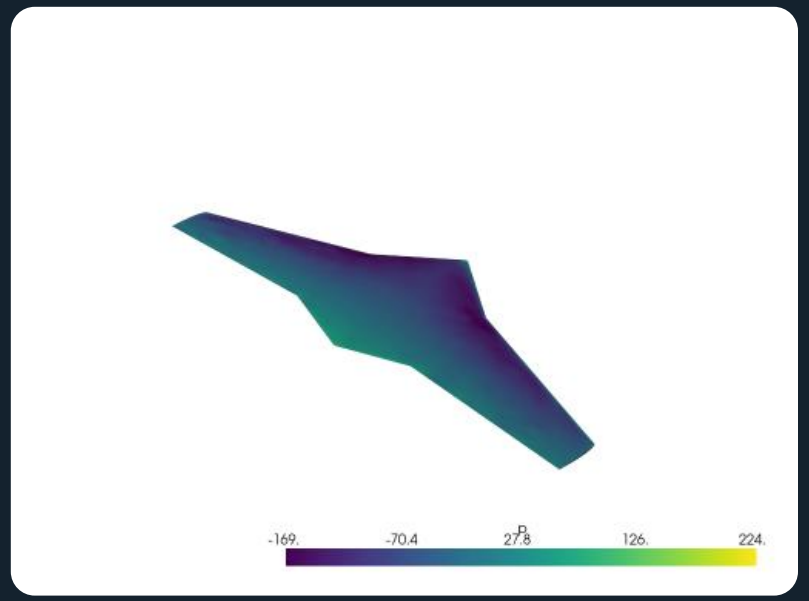
- Use the watertight STL triangles directly as the aircraft boundary.
- Generate the farfield with deterministic Delaunay tetrahedralization.
- Classify aircraft walls and the symmetry plane before case writing.
- Stop on zero cells, conversion failure or a failed mesh-quality gate.

Source: recorded verification report [C]

Flow fields make the result physically inspectable

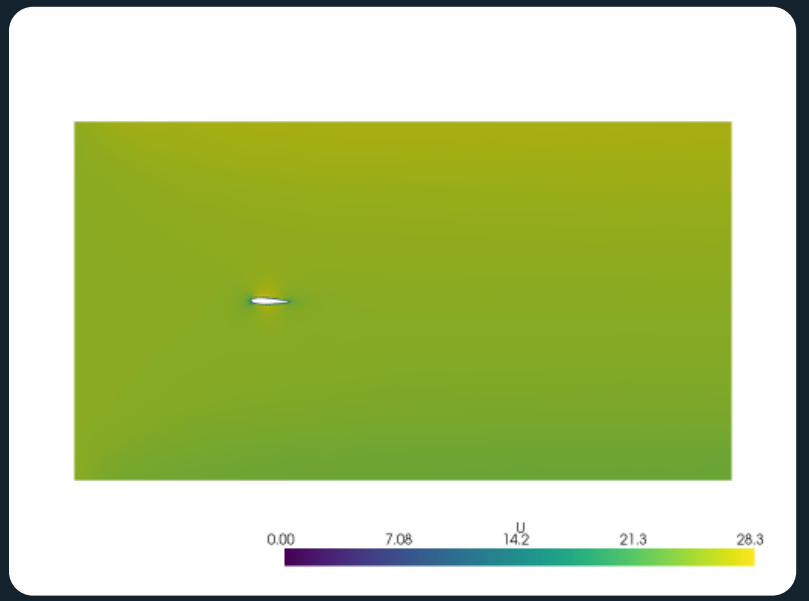
Pressure and velocity images are presented as evidence from the named verification case, not as generic simulation decoration.

SURFACE PRESSURE



The surface field shows the pressure distribution over the BWB, including the leading-edge suction region [C].

VELOCITY MAGNITUDE

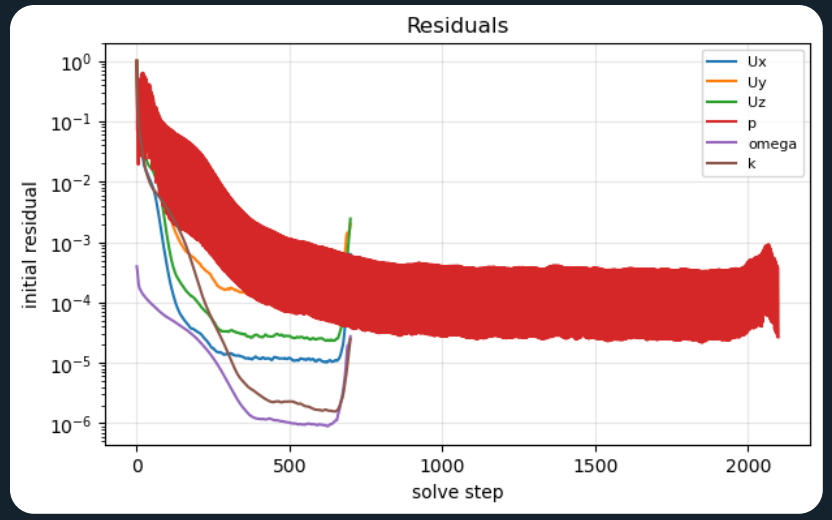


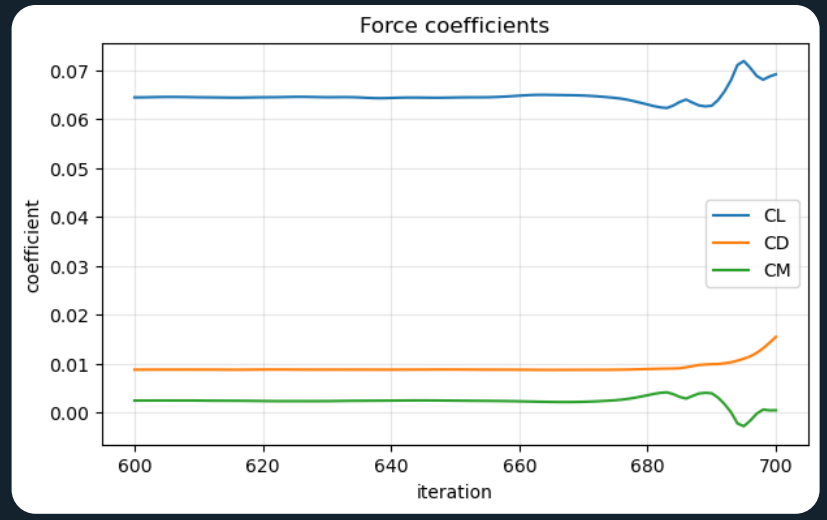
The symmetry-plane slice shows stagnation, acceleration over the body and the downstream wake [C].

VERIFICATION

A clean solver exit is not the engineering verdict

Astra combines convergence, force history, plausibility and the mission target before accepting a result.





WHAT THE EVIDENCE SUPPORTS

The case ran, the monitored solution settled, the fields can be inspected and the quality gate identified a mission-level disagreement.

WHAT REMAINS UNPROVEN

The coarse result is not a validated final aerodynamic performance prediction and does not close mesh independence.

RECORDED RESULT [C]: CFD CL 0.065 VS REQUIRED 0.181 -> REVISE-DESIGN

Source: recorded verification report [C]

ENGINEERING OUTCOMES

The project demonstrates judgement as well as automation

The strongest outcome is a workflow that exposes uncertainty, records decisions and refuses unsupported certainty.

DEMONSTRATED CAPABILITIES

- Requirement decomposition and parametric sizing
- CAD-to-mesh workflow integration
- Multi-fidelity aerodynamic analysis
- Verification gates and failure diagnosis
- Technical UI and engineering communication

ENGINEERING LESSONS

- A successful process exit does not establish valid physics.
- Probabilistic steps need deterministic validators and fallbacks.
- Mesh quality can dominate solver behaviour and result credibility.

NEXT EXPERIMENT [D]

run controlled coarse, medium and fine mesh tiers, then compare force stability, lift and near-wall resolution. The hypothesis is falsified if refinement does not narrow the model disagreement or improve the near-wall evidence.

Source: recorded verification report [C]

EVIDENCE AND LIMITS

Astra is a transparent extension, not a black-box claim

The portfolio separates inherited infrastructure, authored engineering work, measured evidence and unresolved limitations.

ATTRIBUTION

Astra extends the open-source Foam-Agent CFD foundation. The inherited graph remains the CFD spine; the BWB design loop, cockpit, design-session lifecycle and physical-quality gates are the Astra extension.

CURRENT LIMITS

- Aerodynamics-first conceptual design only
- No structural, propulsion or manufacturing validation
- No flight-test evidence
- VLM does not model stall; stall limits are analytic
- The displayed CFD result remains a coarse verification case

TOOLCHAIN

Python / OpenVSP / VSPAERO / Gmsh / OpenFOAM v10 / FastAPI / three.js / LangGraph

Lindon Vaz linkedin.com/in/lindon-vaz- github.com/L-Vaz/Astra

Engineering work should be defendable: source the number, show the problem, record the decision.

Source: README.md